

AJAY BARMAN

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TECHNICAL SKILLS

Programming Languages: JavaScript, TypeScript, Rust, Python (familiar)

Technologies & Frameworks: React.js, Next.js, Node.js, Express.js, TanStack Query, Socket.io, WebRTC, Tailwind CSS, Prisma, Drizzle, Mongoose

Databases & Cloud: PostgreSQL, MongoDB, AWS (EC2), Nginx, Docker, Vercel, Neon, Cloudinary

AI & Developer Tools: AgentKit, Inngest, OpenRouter, Neovim, Postman, Cursor, Git, GitHub

PROJECTS

V0 Clone | Next.js, TanStack Query, Inngest, E2B, AgentKit, OpenAI API, Clerk(OAuth), Prisma, PostgreSQL(Neon) Aug 2026 – Sep 2026

- Built and shipped an agentic AI app-builder that turns plain-language website descriptions into production-ready web apps — generated 500+ apps for 250+ users with a 78% build-success rate and ~4 min median time-to-first-version.
- Engineered a multi-agent build pipeline on Inngest + E2B sandboxes, orchestrating a 4-tool code-agent network (terminal, file read/write, build verification) across up to 10 retry iterations gated by an automated tsc type-check — cut failed builds by ~40%.
- Optimized LLM spend with OpenAI prompt caching plus an "edit prior build" memory model that re-injects prior generated files into fresh sandboxes — reduced per-rebuild token cost ~55% at a ~70% cached-input hit rate.
- Shipped production infra end-to-end: Clerk OAuth, a credit/rate-limit system that throttled LLM calls and curbed abusive API spend by ~30%, GitOps-style CI/CD to Vercel with Inngest Cloud sync, and Neon-hosted Postgres at 99.9% uptime / <50ms p50 query latency.

CodeLeet | Next.js, Clerk(OAuth), Judge0, MonacoEditor, AWS, Nginx, Prisma, PostgreSQL (Neon) May 2026 – Aug 2026

- Architected CodeLeet, a full-stack LeetCode clone in Next.js 16 + TypeScript, with a self-hosted Judge0 engine on AWS EC2 executing code in 6 languages (Python, Java, C++, JavaScript, TypeScript, Rust) across 15 problems (7 Easy / 6 Medium / 2 Hard) — batch-running up to 6 hidden test cases per submission with sub-200ms execution (avg 0.15s, p95 0.16s), replacing paid Judge0 API entirely.
- Built a Monaco-powered IDE with tabbed testcase/result panels (auto-switch, animated run states), per-user progress tracking across 15 problems, expandable submission history, and playlist/bookmark CRUD — gated by server-side Clerk sessions and RBAC (admin-only problem creation enforced in API routes), persisted across 7 relational models in Postgres.
- Hardened and deployed a multi-service stack — Next.js under PM2 behind Nginx (port 80 only), Dockerized Judge0, PostgreSQL, and Redis, Clerk sessions, and Hostinger DNS — with the entire stack (app + sandbox + DB + workers) running on a single 1 vCPU / 1 GB RAM t4g.micro with 30 GB EBS, live in production for under ~\$10/mo.

Threadly | React.js, Turborepo, Express.js, MongoDB, Mongoose, Cloudinary, Socket.io, WebRTC, Mediasoup Feb 2025 – Jul 2026

- Engineered a real-time chat platform ("Threadly") with Socket.io, WebRTC, and a mediasoup SFU, delivering instant messaging, typing indicators, and 1:1 & group video/audio calls across 35 REST endpoints and 22 real-time socket events, supporting 3+ concurrent call participants with low-latency P2P relay and a collaborative canvas whiteboard with live stroke sync and persistence.
- Implemented hardened authentication with JWT access/refresh token rotation (24h/10-day expiry), bcrypt hashing, and httpOnly cookies; defended file handling against SSRF attacks via backend-proxied Cloudinary downloads and applied rate limiting (5,000 requests/15 min per IP) to block abuse and brute-force attempts — reducing security-critical attack surface across the full upload/download pipeline.
- Architected a pnpm + Turborepo monorepo (2 apps, 3 shared packages, ~11,500 LOC) with strict TypeScript across the board, reusable ESLint/tsconfig presets, and a unified dev/build/lint/type-check pipeline, cutting per-feature setup overhead and enabling isolated package builds via Turborepo's task cache.

ACHIEVEMENTS & CERTIFICATIONS

- **Problem Solving:** Solved 300+ problems across LeetCode and NeetCode by applying data structures, algorithm design, and complexity analysis to optimize runtime and memory performance.
- **Certificates:** LLM Engineering, RAG, & AI Agents on Udemy (by Prof. Ryan Ahmed), Complete React and NextJS course with AI powered Projects (by Hitesh Choudhary, Suraj Kumar Jha)
- **Leadership:** Spearheaded a cross-functional team through two consecutive Smart India Hackathons, securing a spot in the top 15% (out of 200+ teams) of competitors in the Inter-College rounds.

EDUCATION

ABES INSTITUTE OF TECHNOLOGY

Bachelor of Technology, Major in Computer Science

Cumulative GPA: 7.5/10.0 (Recent Semester GPA: 8.14/10.0)

Ghaziabad, India

Jul 2023 - Jul 2027

JKG INTERNATIONAL SCHOOL VIJAYNAGAR

CBSE XII PCM

Score: 86.88%

Ghaziabad, India

Apr 2020 - Apr 2022